ASCE 41-23 Tier 1 Checklists

FIRM:	
PROJECT NAME:	
SEISMICITY LEVEL:	
PROJECT NUMBER:	
COMPLETED BY:	
DATE COMPLETED:	
REVIEWED BY:	
REVIEW DATE:	

Legend: C = Compliant, NC = Noncompliant, N/A = Not Applicable, U = Unknown

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Project Name _____
Project Number _____

17.4IO Structural Checklist for Building Types S1: Steel Moment Frames with Stiff Diaphragms and S1a: Steel Moment Frames with Flexible Diaphragms

Table 17-9. Immediate Occupancy Checklist for Building Types S1 and S1a.

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Very Low Seismicity							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	DRIFT CHECK: The drift ratio of the steel moment frames, calculated using the Quick Check procedure of Section 4.4.3.1 is less than 0.015.	5.5.2.1.2	A.3.1.3.1	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	COLUMN AXIAL STRESS CHECK: The axial stress caused by gravity loads in columns subjected to overturning forces is less than $0.10F_y$. Alternatively, the axial stress caused by overturning forces alone, calculated using the Quick Check procedure of Section 4.4.3.6 is less than $0.30F_y$.	5.5.2.1.3	A.3.1.3.2	
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	FLEXURAL STRESS CHECK: The average flexural stress in the moment-frame columns and beams, calculated using the Quick Check procedure of Section 4.4.3.9, is less than F_y. Columns need not be checked if the strong column–weak beam checklist item is compliant.	5.5.2.2.2	A.3.1.3.3	
Connections							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation with a minimum of two anchor rods and with the base plates bearing on concrete or a grout pad. The anchor rods are capable of resisting the overturning force using the Quick Check procedure of Section 4.4.3.6.	5.7.3.1	A.5.3.1	

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
Low Seismicity (Complete the Following Items in Addition to the Items for Very Low Seismicity)							
Seismic-Force-Resisting System							
C ☐	**NC** ☐	**N/A** ☐	**U** ☐	REDUNDANCY: The number of lines of moment frames in each principal direction is greater than or equal to 2. The number of bays of moment frames in each line is greater than or equal to 3.	5.5.1.1	A.3.1.1.1	

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Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
C	NC	N/A	U	INTERFERING WALLS: All concrete and masonry infill walls placed in moment frames are isolated from structural elements.	5.5.2.1.1	A.3.1.2.1	
Connections							
C	NC	N/A	U	TRANSFER TO STEEL FRAMES: Diaphragms are connected for transfer of seismic forces to the steel frames, and the connections are able to develop the lesser of the strength of the frames or the diaphragms.	5.7.2	A.5.2.2	
C	NC	N/A	U	STEEL COLUMNS: The columns in seismic-force-resisting frames are anchored to the building foundation, and the anchorage is able to develop the least of the following: the tensile capacity of the column, the tensile capacity of the lowest-level column splice (if any), or the uplift capacity of the foundation.	5.7.3.1	A.5.3.1	
Moderate Seismicity (Complete the Following Items in Addition to the Items for Very Low and Low Seismicity)							
Seismic-Force-Resisting System							
C	NC	N/A	U	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the expected strength of the adjoining members based on the specified minimum yield stress of the steel.	5.5.2.2.1	A.3.1.3.4	
C	NC	N/A	U	PANEL ZONES: All panel zones have the shear capacity to resist the shear demand required to develop 0.8 times the sum of the flexural strengths of the girders framing in at the face of the column.	5.5.2.2.3	A.3.1.3.5	
C	NC	N/A	U	COLUMN SPLICES: All column splice details located in moment frames include connection of both flanges and the web, and the splice develops the strength of the column.	5.5.2.2.4	A.3.1.3.6	
C	NC	N/A	U	STRONG COLUMN—WEAK BEAM: The percentage of strong column–weak beam joints in each story of each line of moment-resisting frames is greater than 50%.	5.5.2.1.5	A.3.1.3.7	

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C	NC	N/A	U			
☐	☐	☐	☐	COMPACT MEMBERS: All frame elements meet section requirements in accordance with AISC 341, Table D1.1a, for highly ductile members.	5.5.2.2.5	A.3.1.3.8
☐	☐	☐	☐	BEAM PENETRATIONS: All openings in frame-beam webs are less than one-quarter of the beam depth and are located in the center half of the beams.	5.5.2.2.6	A.3.1.3.9
☐	☐	☐	☐	GIRDER FLANGE CONTINUITY PLATES: There are girder flange continuity plates at all moment-frame joints.	5.5.2.2.7	A.3.1.3.10
☐	☐	☐	☐	OUT-OF-PLANE BRACING: Beam–column joints are braced out of plane.	5.5.2.2.8	A.3.1.3.11
☐	☐	☐	☐	BOTTOM FLANGE BRACING: The bottom flanges of beams are braced out of plane.	5.5.2.2.9	A.3.1.3.12
Diaphragms (Stiff or Flexible)						
☐	☐	☐	☐	PLAN IRREGULARITIES: There is tensile capacity to develop the strength of the diaphragm at reentrant corners or other locations of plan irregularities.	5.6.1.4	A.4.1.7
☐	☐	☐	☐	DIAPHRAGM REINFORCEMENT AT OPENINGS: There is reinforcing around all diaphragm openings larger than 50% of the building width in either major plan dimension.	5.6.1.5	A.4.1.8
☐	☐	☐	☐	OPENINGS AT FRAMES: Diaphragm openings immediately adjacent to the moment frames extend less than 15% of the total frame length.	5.6.1.3	A.4.1.5
Flexible Diaphragms						
☐	☐	☐	☐	CROSSTIES: There are continuous crossties between diaphragm chords.	5.6.1.2	A.4.1.2
☐	☐	☐	☐	STRAIGHT SHEATHING: All straight-sheathed diaphragms have horizontal spans less than 12 ft (3.6 m) and aspect ratios less than 1-to-1 in the direction being considered.	5.6.2	A.4.2.1
☐	☐	☐	☐	DIAGONALLY SHEATHED AND UNBLOCKED DIAPHRAGMS: All diagonally sheathed or unblocked wood structural panel diaphragms have horizontal spans less than 30 ft (9.2 m) and aspect ratios less than or equal to 3-to-1.	5.6.2	A.4.2.2

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C	NC	N/A	U			
☐	☐	☐	☐	NON-CONCRETE-FILLED DIAPHRAGMS: Bare steel deck diaphragms or steel deck diaphragms with fill other than reinforced concrete consist of horizontal spans of less than 40 ft (12.2 m) and have aspect ratios less than 4-to-1.	5.6.3	A.4.3.1
C	**NC**	**N/A**	**U**	OTHER DIAPHRAGMS: Diaphragms do not consist of a system other than wood, steel deck, concrete, or horizontal bracing.	5.6.5	A.4.7.1
☐	☐	☐	☐			
Foundation System						
C	**NC**	**N/A**	**U**	OVERTURNING: The ratio of the least horizontal dimension of the seismic-force-resisting system at the foundation level to the building height (base/height) is greater than $0.6S_a$.	5.4.3.3	A.6.2.1
☐	☐	☐	☐			

Status				Evaluation Statement	Tier 2 Reference	Commentary Reference	Comments
High Seismicity (Complete the Following Items in Addition to the Items for Very Low, Low, and Moderate Seismicity)							
Seismic-Force-Resisting System							
C	**NC**	**N/A**	**U**	MOMENT-RESISTING CONNECTIONS: All moment connections are able to develop the strength of the adjoining members or panel zones based on 110% of the expected yield stress of the steel in accordance with AISC 341, Section A3.2.	5.5.2.2.1	A.3.1.3.4	
☐	☐	☐	☐				

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